

Antenna Theory and Design

Chapter 1: Introduction, Radiation Concepts, Antenna Parameters, and Antenna Types

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Learning objectives

- Explain the role of antennas in wireless communication, radar, and sensing.
- Interpret an antenna as a transition between a transmission line and free space.
- Describe radiation using the physical idea of accelerated charge.
- Identify the main antenna parameters: pattern, directivity, gain, polarization, impedance, bandwidth, and scanning.
- Classify antennas into four basic families.
- Work through a numerical half-wave dipole example at 2.4 GHz.



Chapter 1 roadmap

- 1 Historical motivation
- 2 What is an antenna?
- 3 Transmission line versus antenna
- 4 How antennas radiate
- 5 Antenna performance parameters
- 6 Four antenna types
- 7 Gain-based antenna selection
- 8 Numerical example

Central idea

An antenna is not just a metal object; it is a physical, circuit, and spatial energy-conversion system.



Historical motivation: from Maxwell to modern wireless

Year/Period	Milestone
1844	Telegraph: beginning of electronic communication
1864	Maxwell's equations: theoretical foundation for electromagnetic waves
1887	Hertz demonstrates electromagnetic waves and the first antenna experiments
1897	Marconi develops practical wireless radio systems
1901	First transatlantic radio transmission
World War II	Radar growth; horn, reflector, and array antennas become critical
1960s	Satellite communications and optical fiber mature
1980s onward	Cellular and personal wireless communication become widespread

Teaching point

Antenna engineering evolved with communication, radar, microwave devices, and computational electromagnetics.



What is an antenna?

IEEE-style definition

An antenna is the part of a transmitting or receiving system designed to radiate or receive electromagnetic waves.

Guided wave $\longleftrightarrow$ Free-space wave

- Transmit mode: converts a bound wave on a line into a radiated wave.
- Receive mode: converts an incident free-space wave into a guided signal.
- It is the interface between RF circuitry and space.



Transmission line versus antenna

Transmission line

- Confines electromagnetic energy
- Uses conductors or guiding structures
- Goal: avoid radiation
- Example: coax, waveguide, microstrip line

Antenna

- Encourages radiation into space
- Needs no guiding structure after launch
- Goal: controlled radiation
- Example: dipole, patch, horn, reflector

No radiation → controlled radiation



Wavelength is the design scale

Free-space wavelength

$$\lambda = \frac{c}{f}$$

$$\lambda(\text{m}) = \frac{300}{f(\text{MHz})}$$

$$\lambda(\text{cm}) = \frac{30}{f(\text{GHz})}$$

- Antenna size must be compared with λ , not only meters or centimeters.
- Low frequency means large wavelength and physically large antennas.
- GHz frequencies enable compact patches, horns, and arrays.
- Array element spacing is often near $\lambda/2$.



When do we use antennas?

- Mobile systems: aircraft, spacecraft, vehicles, ships, and phones.
- Broadcasting: one transmitter serves many receivers.
- Long-distance links where cable loss or right-of-way cost is prohibitive.
- Radar and active remote sensing: transmit energy and receive scattered returns.
- Passive sensing and radiometry: receive naturally emitted energy.
- Industrial RF systems: heating, drying, and microwave processing.

System-level decision

The choice between cable and radio depends on distance, frequency, mobility, cost, reliability, bandwidth, and security.



Physical mechanism of radiation

Key principle

Accelerated charges radiate electromagnetic energy.

- In an antenna, charges oscillate because of an RF excitation.
- Oscillation means changing velocity and direction, hence acceleration.
- The disturbance in the electromagnetic field detaches and propagates outward.
- Electrons do not leave the antenna and fly through space; the fields propagate.

$$E, H \propto \frac{1}{r}, \quad S \propto \frac{1}{r^2}$$



From open transmission line to dipole

Open-circuited line

- Currents on the two conductors are opposite.
- External fields largely cancel.
- Energy remains guided near the line.

Bent-out conductors

- Currents no longer cancel externally.
- Fields are exposed to free space.
- Radiation becomes possible.

guided structure → radiating dipole



Antenna performance parameters

Parameter	Meaning
Radiation pattern $F(\theta, \phi)$	Angular variation of radiation around the antenna
Directivity D	Concentration of radiation relative to average radiation
Gain G	Directivity reduced by antenna losses; $G = e_r D$
Polarization	Time behavior and orientation of the radiated electric-field vector
Input impedance Z_A	Ratio of terminal voltage to current; key to matching
Bandwidth	Frequency range over which required performance is acceptable
Scanning	Movement of the radiation pattern in space

Design reality

Improving one parameter often degrades another. Antenna design is a trade-off problem.



Radiation pattern and directivity

Radiation pattern

$$F(\theta, \phi)$$

- Main beam
- Side lobes
- Nulls
- Omnidirectional or directive behavior

Directivity

$$D = \frac{\text{peak radiation intensity}}{\text{average radiation intensity}}$$

- Pattern-only quantity
- Does not include losses
- Higher D means narrower spatial concentration



Gain, efficiency, and power interpretation

Core relation

$$G = e_r D$$

$$0 \leq e_r \leq 1$$

- D : spatial concentration
- e_r : radiation efficiency
- G : useful radiating performance

If $P_{in} = 1 \text{ W}$

$$P_{rad} = e_r P_{in}$$

If $e_r = 0.90$, then

$$P_{rad} = 0.90 \text{ W.}$$



Polarization, impedance, bandwidth, and scanning

- **Polarization:** linear, circular, or elliptical behavior of the electric-field vector.
- **Input impedance:**

$$Z_A = R_A + jX_A$$

determines matching to the feed line.

- **Bandwidth:** frequency range over which pattern, gain, impedance, polarization, or efficiency remains acceptable.
- **Scanning:** motion of the beam by mechanical rotation or electronic phase control.

Important distinction

Antenna bandwidth is not only an S_{11} number. It must be tied to a performance requirement.



The four basic antenna types

Type	Key property	Examples
Electrically small	Size much less than λ ; low efficiency and high reactance	short dipole, small loop
Resonant	Good operation near one frequency or narrow bands	half-wave dipole, patch, Yagi-Uda
Broadband	Acceptable pattern, gain, and impedance over wide frequency range	spiral, biconical, log-periodic
Aperture	Physical opening or effective aperture controls radiation	horn, reflector, lens

Quick habit

When you see a new antenna, first ask: which family does it belong to?

Electrically small and resonant antennas

Electrically small

$$L \ll \lambda$$

- Low directivity
- Low radiation resistance
- High reactance
- Often low efficiency

Resonant

$$L \approx \frac{\lambda}{2} \quad \text{or related resonance}$$

- Low to moderate gain
- Easier impedance matching
- Narrow bandwidth
- Examples: dipole, patch, Yagi



Broadband and aperture antennas

Broadband

- Designed for wide frequency ranges
- Often has an active region that shifts with frequency
- Examples: spiral, biconical, log-periodic

Aperture

- Uses an opening or reflecting surface
- High gain and narrow beams
- Gain increases as aperture becomes large relative to wavelength
- Examples: horn, reflector, lens

larger electrical aperture $\Rightarrow$ narrower beam and higher gain



Choosing antenna type by gain target

Approximate gain target	Likely antenna choices
≤ 5 dB	small antennas, loops, dipoles, monopoles
5 to 8 dB	microstrip patches, planar frequency-independent antennas
8 to 15 dB	Yagi-Uda, axial-mode helix, log-periodic dipole array
> 15 dB	aperture antennas, horns, reflectors, or arrays

Design implication

If a system needs 20 dB gain, a single patch or dipole is usually not enough.



Numerical example: half-wave dipole at 2.4 GHz

Problem statement

Consider a thin, center-fed, vertical half-wave dipole in free space operating at

$$f_0 = 2.4 \text{ GHz.}$$

Assume a 50Ω RF system and radiation efficiency

$$e_r = 0.90.$$

Determine wavelength, length, pattern behavior, directivity, gain, polarization, impedance match, and bandwidth.



Example: wavelength and physical length

Wavelength

$$\lambda = \frac{c}{f_0}$$

$$\lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m}$$

$$\lambda = 12.5 \text{ cm}$$

Practical dipole length

Ideal:

$$L \approx \frac{\lambda}{2} = 6.25 \text{ cm}$$

Practical:

$$L \approx 0.48\lambda \approx 6.0 \text{ cm}$$



Example: pattern, directivity, and gain

Pattern behavior

$$F(\theta) = \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta}$$

- Maximum at $\theta = 90^\circ$
- Nulls at 0° and 180°
- Doughnut-shaped 3D pattern
- $HPBW \approx 78^\circ$

Directivity and gain

$$D \approx 1.64 = 2.15 \text{ dBi}$$

$$G = e_r D = 0.90(1.64) = 1.476$$

$$G \approx 1.69 \text{ dBi}$$



Example: polarization and input impedance

Polarization

A vertical dipole produces

linear vertical polarization.

If the receive antenna is tilted by ψ ,

$$PLF = \cos^2(\psi).$$

For $\psi = 45^\circ$:

$$PLF = 0.5 = -3 \text{ dB}.$$

Input impedance

At resonance, approximately

$$Z_A \approx 73 + j0 \ \Omega.$$

For a $50 \ \Omega$ system:

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0}.$$



Example: mismatch calculations

Reflection coefficient

$$\Gamma = \frac{73 - 50}{73 + 50} = \frac{23}{123} = 0.187$$

$$|\Gamma| = 0.187$$

$$RL = -20 \log_{10} |\Gamma| = 14.6 \text{ dB}$$

VSWR and accepted power

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 1.46$$

$$P_{acc} = P_{inc}(1 - |\Gamma|^2)$$

For $P_{inc} = 1 \text{ W}$:

$$P_{acc} \approx 0.965 \text{ W}$$

Mismatch loss $\approx 0.15 \text{ dB}$.



Example: bandwidth and summary

Assumed impedance bandwidth

Criterion:

$$S_{11} < -10 \text{ dB}$$

Suppose:

$$f_L = 2.34 \text{ GHz}, \quad f_H = 2.46 \text{ GHz}$$

$$BW = f_H - f_L = 120 \text{ MHz}$$

$$FBW = \frac{BW}{f_0} = 5\%$$

Quantity	Value
λ	12.5 cm
L	≈ 6.0 cm
Directivity	2.15 dBi
Gain	1.69 dBi
Input impedance	$73 + j0 \Omega$
Return loss	14.6 dB
VSWR	1.46
Bandwidth	120 MHz



Connection to radar, arrays, and DOA

Real array response

Measured array response = element pattern $\times$ array factor

- Element pattern changes gain with angle.
- Polarization mismatch changes received amplitude.
- Impedance mismatch reduces accepted power.
- Mutual coupling can distort the array manifold.
- Bandwidth limits wideband radar and communication performance.

Practical lesson

A MUSIC or beamforming simulation with ideal isotropic sensors can be much cleaner than real hardware.

Final takeaways

- 1 An antenna converts guided waves and free-space waves.
- 2 Radiation is produced by time-varying currents and accelerated charges.
- 3 Wavelength is the main physical scale in antenna engineering.
- 4 Pattern, directivity, gain, polarization, impedance, bandwidth, and scanning describe antenna performance.
- 5 Antenna choice depends on required gain, bandwidth, size, frequency, and system use.
- 6 In radar and DOA systems, antenna physics shapes the signal-processing problem.

Suggested classroom activity

Ask students to identify one example of each antenna family in the real world and estimate its operating frequency range.



- W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed., Wiley, Chapter 1.
- Numerical dipole example uses standard half-wave dipole approximations introduced in antenna fundamentals.

